

Multi-View, Multi-Scale, Geometrically-Consistent Multi-View Stereo (Supplementary Materials)

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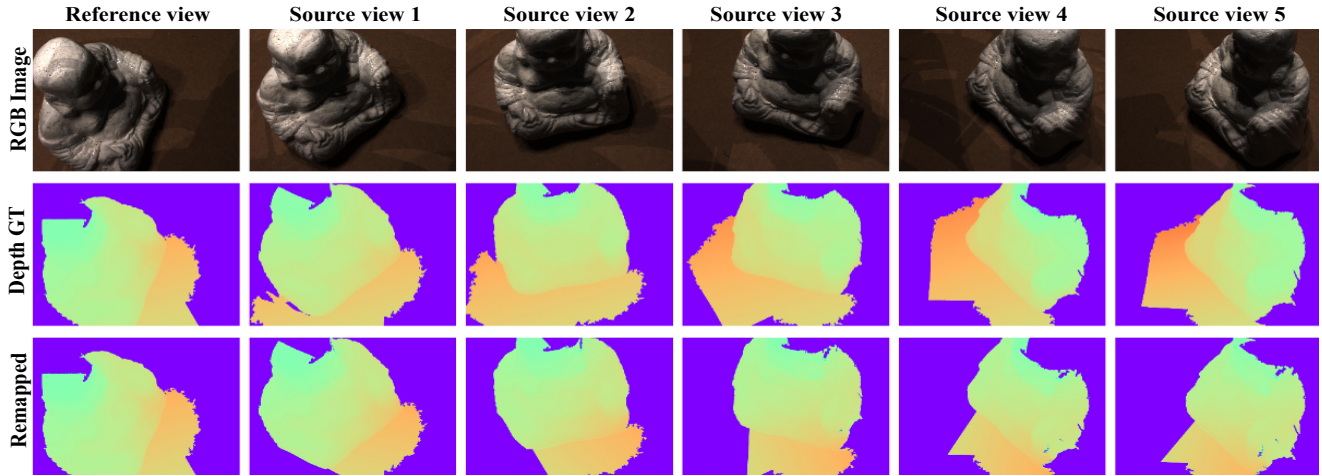


Figure 1: First row shows the selection of M closest source images for a given reference image. Middle row shows the corresponding ground truth depth maps and last row shows the remapped source ground truth depth maps using x-y coordinates of reference view projection to the source view. During remapping, all additional pixels from the source views are ignored. The remapped depths are then back-projected to source view to generate mask. Finally, reference view mask is applied on per-pixel penalty to restrict the penalties. Corresponding final ξ_p is shown in Fig. 3 of the paper. All depth maps are shown within respective view mask.

A. Occlusion and its impact

Modeling occlusion of pixels in multi-view setting is a difficult problem. It is difficult to reason about a pixel in a view whose corresponding 3D points are occluded in other view. The problem becomes significant if a penalty is being attached to all such pixels, like in the proposed multi-view geometric consistency checking module. The GC module checks geometric consistency of each pixel across multiple source views and awards a penalty for inconsistency. Assigning penalties to occluded pixels and multiplying it with depth error adversely impacts the training process. Early in our experiments, we observe that the loss started to explode with training, i.e. as the model trains the loss values starts to increase.

Our investigation suggests that the wrongful penalties of occluded pixels dominated loss during training. We find that our method becomes robust to this problem with a se-

ries of steps taken. First, we use the closest source view images as defined by MVSNet [16]. The first row of Fig. 1 shows the source view selection for the given reference view. Choosing closest view to the reference view reduces the number of possible occluded pixels. Second, during forward-backward-reprojection, we remap the source view depth map as per the x-y coordinate projections of the reference view to the source view and then, the remapped values are back-projected to the reference view (see Alg. 2 in the paper). The last row in Fig. 1 shows the remapped version of the source view depth maps. During remapping, all the occluded as well as the additional pixels of the source view is dropped and then this remapped version is back-projected. This handles the extreme cases of occlusion or additional visible pixels. At the end, once the per-pixel penalty is generated, we apply the reference view binary mask on it to do away with any such pixel which is not part of the scene in consideration (see Fig. 3 in the paper).

The combination of these steps help us control the impact of wrongful penalties and stabilize the training process.

B. Geometric Consistency Module

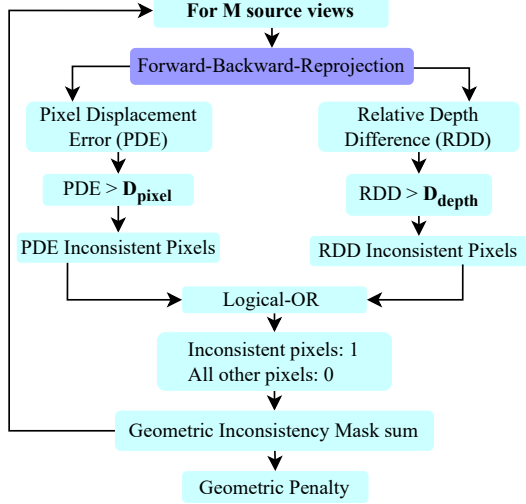


Figure 2. GC module flow-chart for consistency check.

We describe the steps of geometric consistency (GC) module in Fig. 2. At each stage, the geometric consistency of estimated depth map is checked across M source views. For each source view, we perform the forward-backward-reprojection of estimated depth map to reason about geometric inconsistency of pixels (described in Alg. 2). In this three-step process, first, we warp each pixel P_0 of a reference view depth map D_0 to its i^{th} neighboring source view to obtain corresponding pixel P'_i . Then, we back-project P'_i into 3D space and finally, we reproject it to the reference view as P''_0 using c_0 . D_0 , $D'_{P'_i}$ and $D''_{P''_0}$ represents depth value of pixels associated with P_0 , P'_i and P''_0 [6]. With P_0 and $D''_{P''_0}$, we calculate pixel displacement error (PDE) and relative depth difference (RDD). After taking logical-OR between PDE and RDD, we assign value 1 to all inconsistent pixel and zero to all other pixels. The geometric inconsistency mask sum is generated over M source views and averaged to generate per-pixel penalty ξ_p .

C. Depth Interval Ratio (DIR)

ξ_p Range	Stage-wise DIR	Acc↓	Comp↓	Overall↓
[1, 3]	2.0, 0.8, 0.40	0.338	0.269	0.3035
[1, 3]	2.0, 0.7, 0.35	0.343	0.264	0.3035
[1, 3]	2.0, 0.7, 0.30	0.331	0.27	0.3005
[1, 3]	1.6, 0.7, 0.30	0.329	0.271	0.300

Table 1. The performance of GC-MVSNet on evaluation set of DTU [8] with change in stage-wise DIR (depth interval ratio).

DIR directly impacts the separation of two hypothesis planes at pixel level. For a given stage, the pixel-level depth interval is calculated as product of DIR_{stage} and *depth interval* (DI). The value of DI is calculated using *interval scale* and a constant value provided in DTU camera parameter files.

Following the trend of modern learning-based methods [1, 3, 5, 10, 15, 16, 19], we train our model on 512×640 resolution and test on 864×1152 resolution. To adjust for the pixel-level depth interval caused by the increase in resolution, we explore different DIR values for testing on DTU. We train our model with stage-wise DIR 2.0, 0.8, 0.4 (DIR_{train}), such that the refine stage pixel-level depth interval is same as the provided *interval scale* value of 1.06. Table 1 shows DIR values for evaluation on DTU, we only explore smaller values than DIR_{train} to compensate for the increase in resolution. GC-MVSNet achieves its optimal performance at DIR 1.6, 0.7, 0.3 with $\xi_p \in [1, 3]$, DIR_{test} . We use the same DIR_{train} and DIR_{test} with $\xi_p \in [1, 2]$.

D. Stabilizing the Training Process

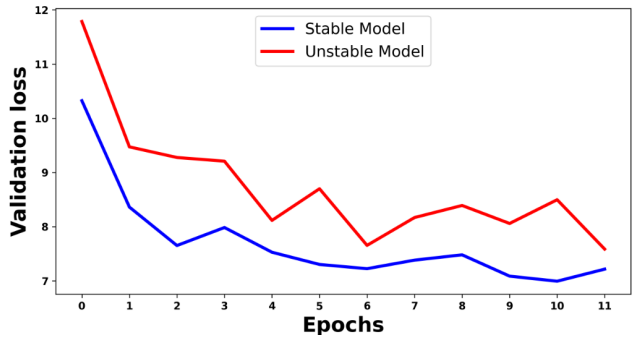


Figure 3. Validation loss on DTU [8] dataset during training. The red line shows the unstable model training, validation loss change in zig-zag manner. Blue line shows stable training with smooth change in validation loss.

Most of the modern learning-based MVS methods [3, 5, 10, 12, 13, 20] use BatchNorm [7] along with Apex (Nvidia) for batch synchronization. BatchNorm is most useful with large batch size. For smaller batch size, like 1 or 3, it degrades the training process [7] by poor estimation of population mean (μ) and std. (σ) over small batch size.

GroupNorm [14] alleviates this problem by estimating μ and σ along the channels instead of batch. Weight-standardization [11] further stabilizes the training and evaluation steps. We refer to the original papers for further understanding of these concepts. GC-MVSNet replaces BatchNorm with GroupNorm and Weight-standardization techniques to stabilize the training process. Fig. 3 shows the difference between model trained with (red line) and without (blue line) BatchNorm. With the use of GroupNorm along with Weight-standardization, the evaluation

loss curve become smooth and stable.

E. Depth Map Fusion Methods

The quality of point clouds depends heavily on depth fusion methods and their hyperparameters. Following the recent learning-based methods [3, 5, 10], we also use different fusion method for DTU and Tanks and Temples dataset. For DTU, we use Fusibile [4] and for Tanks and Temples, we use Dynamic method [3, 12].

Fusibile fusion method uses three hyperparameters, disparity threshold, probability confidence threshold, and consistency threshold. Disparity threshold defines the upper limit of disparity for points to be eligible for fusion. Probability confidence threshold defines the lower limit of confidence above which points are eligible for fusion. The consistency threshold mandates that the eligible points be geometrically consistent across as many source views. During the fusion process, only those points that satisfy all three conditions are fused into point cloud.

Dynamic fusion method uses only two hyperparameters, probability confidence threshold and consistency threshold. Both these hyperparameters have exact same function as in Fusibile method. The disparity threshold is not provided by the user, it is dynamically adjusted during the fusion process.

F. Accuracy and Completeness Metrics

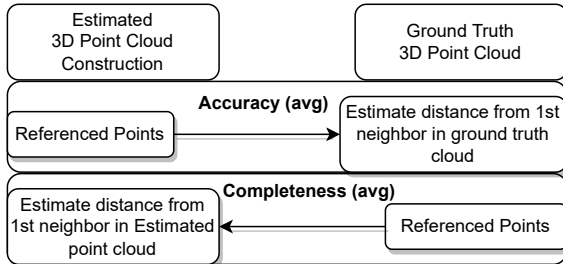


Figure 4. The process of calculating accuracy and completeness for DTU [8] point cloud evaluation.

Accuracy and completeness are two metrics used with DTU [8] dataset. Fig. 4 shows the process of calculation. Accuracy is the average of the distance of the first neighbor from predicted point cloud to ground truth point cloud. It only considers the points which are below the maximum threshold for the distance. For completeness, same process is repeated but with ground truth as referenced point cloud, i.e. the average of the distance of the first neighbor from the ground truth point cloud to the predicted point cloud.

G. Use of Existing Assets

We use PyTorch to implement GC-MVSNet. It is based on CasMVSNet [5] and TransMVSNet [3]. These two

methods heavily borrow code from the PyTorch implementation of MVSNet [16].

We use preprocessed images and camera parameters of DTU [8] dataset from official repository of MVSNet [16] and R-MVSNet [17]. We follow [2] for training and testing on BlendedMVS [18]. For Tanks and Temples [9] evaluation, we use images and camera parameters as used in R-MVSNet [17].

H. Point Clouds

In this section, we show all evaluation set points clouds reconstructed using GC-MVSNet on DTU [8], Tanks and Temples [9] and BlendedMVS [18] datasets. Fig. 5, 6 and 7 show all evaluation set point clouds from DTU, Tanks and Temples and BlendedMVS, respectively.



Figure 5. Point clouds reconstructed using GC-MVSNet for all scenes from DTU [8] evaluation set.

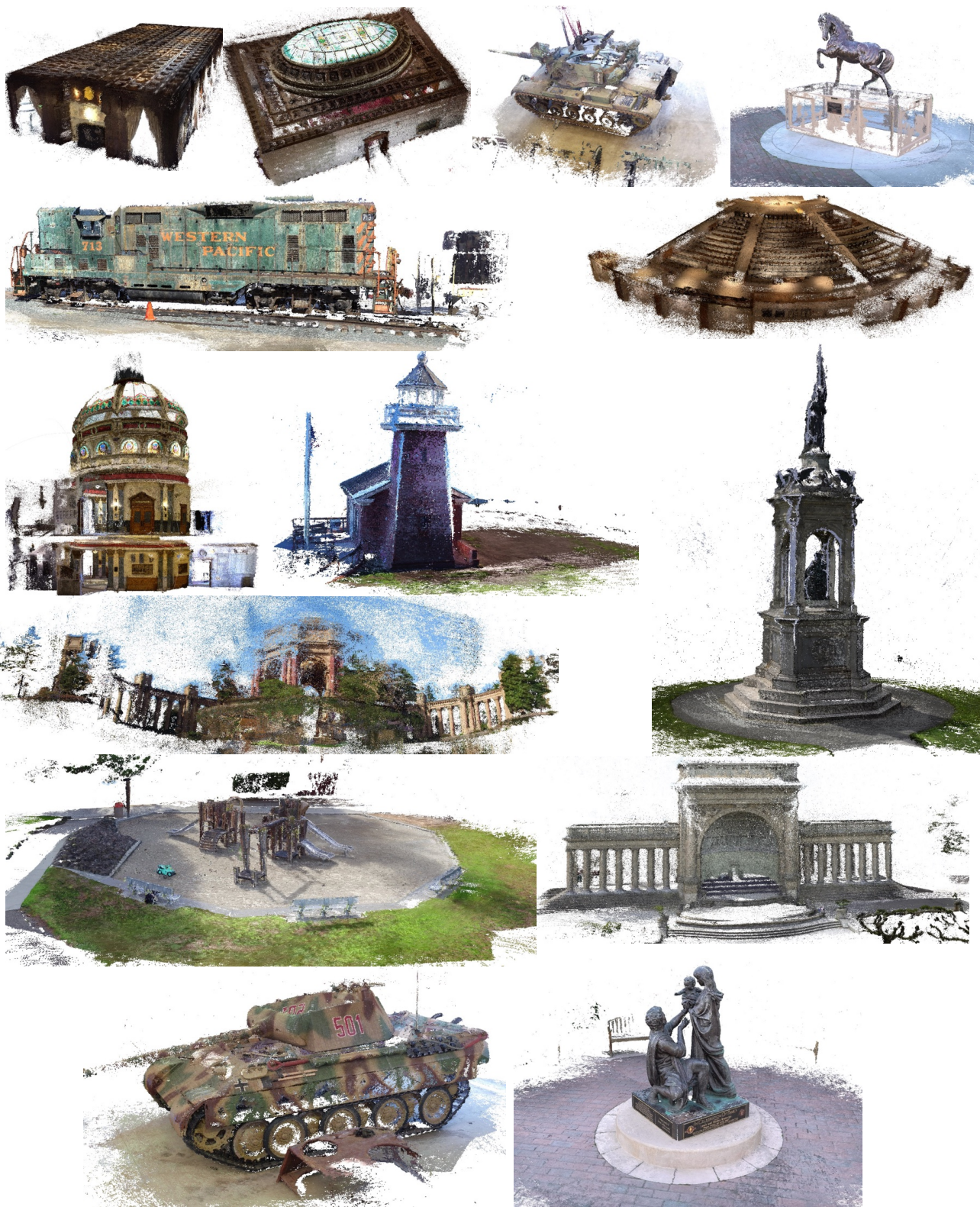


Figure 6. Point clouds reconstructed using GC-MVSNet for all scenes from Tanks and Temples [9] intermediate and advanced set.



Figure 7. Point clouds reconstructed using GC-MVSNet for all scenes from BlendedMVS [18] evaluation set.

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